Continuous Convergence and Curried Functional Spaces

1. Introduction

2. General curried functions

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined recursively as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we let $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise linear structure and the topology of pointwise convergence (see Section 4.3 of the Appendix). Lemma 4.3.1 and Lemma 4.3.2 ensure that $H_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

Definition 2.2. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma\left(\tilde{f}\right)$ will be called the *curried version* of $\tilde{f}$.

Lemma 2.1. *For every $n \in \mathbb{N}_0$, the space $H_n(\mathbb{R})$ is locally compact.*

Proof: An easy proof by induction over n. ■

Lemma 2.2. *For every $n \geq 1$, the space $H_n(\mathbb{R})$ is not metrizable.*

Proof:

■

3. Continuous curried functions

Definition 3.1. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, endowed with pointwise linear structure and the compact-open topology (see Section 4.4 of the Appendix).
Lemma 4.4.1 and Lemma 4.4.2 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Lemma 3.1. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

Proof: First, let F be the derived filter of $\{f_k\}_{k\in\mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k\in\mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By Proposition 4.4.1, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0,m_0} = \{f_n(x_m) \mid n \geq n_0,\ m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

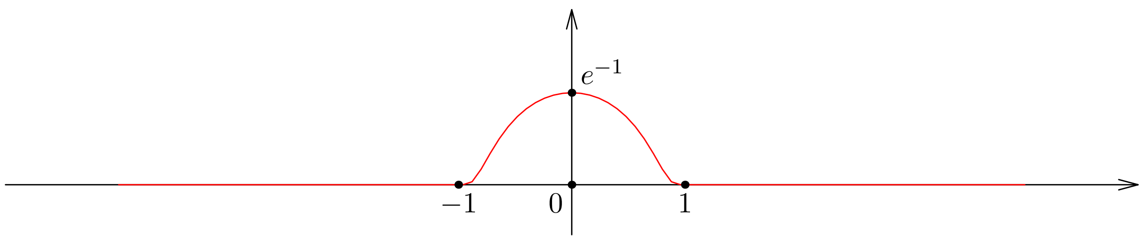
Now, to show that $f_k \to f$ in $\mathcal{C}(X,Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m\in\mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_m}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n\to\infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n\to\infty]{} f(x)$ by (1), a contradiction. ∎

Corollary 3.1. *From the above lemma we conclude that if $n \geq 1$, a sequence of functions $\{f_k\}_{k=1}^{\infty}$ in $C_n(\mathbb{R})$ converges to $f \in C_n(\mathbb{R})$ if and only if for any sequence $\{x_k\}_{k=1}^{\infty} \subset \mathbb{R}$ converging to $x \in \mathbb{R}$, we have $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ in $C_{n-1}(\mathbb{R})$.*

Example 3.1. The topology on $C_n(\mathbb{R})$ is finer than the topology induced from $H_n(\mathbb{R})$, and does not coincide with it, unless $n = 0$. To see this, take $n = 1$ and consider the classical function

$$\varphi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2-1}\right), & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

This is a $C^\infty(\mathbb{R})$ function supported on $[-1, 1]$. Its graph looks as follows:



Now consider the following sequence of functions:

$$f_k(x) = e \cdot \varphi\Big(k \cdot \left(x - \frac{1}{k}\right)\Big).$$

Clearly, f_k is infinitely differentiable on $\mathbb{R}$, supported on $[0, 2/k]$, and has a maximum at $x = 1/k$ with $f_k(1/k) = 1$. It is clear that for all $x \in \mathbb{R}$, we have $f_k(x) \xrightarrow[k\to\infty]{} 0$, so $f_k \to 0$ in the topology induced from $H_1(\mathbb{R})$. However, f_k do not converge to 0 in the native topology of $C_1(\mathbb{R})$. To see this, take $x_k = 1/k$ and observe that $f_k(x_k) \equiv 1 \neq 0$.

A similar argument applies for any other $n \geq 2$. Hence we conclude that $C_n(\mathbb{R})$ is not a subspace of $H_n(\mathbb{R})$.

Proposition 3.1. *Let $n \in \mathbb{N}_0$. Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(x_1)(x_2)...(x_n) = f(x_1, ..., x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 4.4.2, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ∎

Lemma 3.2. *For every $n \in \mathbb{N}_0$, the space $C_n(\mathbb{R})$ is locally compact.*

Proof: An easy proof by induction over n. ∎

3.1. Differentiable functions

Definition 3.1.1. Let $m \in \mathbb{N}_0$ and $n \in \mathbb{N}_0$. The set $D_n^m(\mathbb{R})$ of *curried m times differentiable functions* is a topological space defined recursively as follows:

- If $n = 0$, we set $D_0^m(\mathbb{R}) = \mathbb{R}$ for all m, as usual;
- If $m = 0$, we set $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$;
- Let $m, n > 0$, and assume that both $D_{n-1}^m(\mathbb{R})$ and $D_n^{m-1}(\mathbb{R})$ are defined. Then we let $D_n^m(\mathbb{R})$ be the set of all functions $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ such that there is a function $f' \in D_n^{m-1}(\mathbb{R})$ which satisfies

$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h\to 0]{} f'(x) \in H_{n-1}(\mathbb{R})$$

for all $x \in \mathbb{R}$, where **the convergence is taken in the space $H_{n-1}(\mathbb{R})$**. Clearly, if $f, g \in D_n^m(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)' = \alpha f' + \beta g',$$

so in particular, $\alpha f + \beta g \in D_n^m(\mathbb{R})$.

We endow $D_n^m(\mathbb{R})$ with the coarsest topology in which the maps

$$\begin{aligned} i : D_n^m(\mathbb{R}) &\to \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), \qquad & d : D_n^m(\mathbb{R}) &\to D_n^{m-1}(\mathbb{R}) \\ f &\mapsto f & f &\mapsto f' \end{aligned} \tag{2}$$

are continuous. This topology will be called the *differential topology of order m.* We prove below that $D_n^m(\mathbb{R})$ are Hausdorff topological vector spaces.

Lemma 3.1.1. *For all $m, n \in \mathbb{N}_0$, we have $D_n^{m+1}(\mathbb{R}) \subset D_n^m$, and the inclusion map*

$$j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$$

is continuous, meaning that the topology on $D_n^{m+1}(\mathbb{R})$ is finer than the topology induced from $D_n^m(\mathbb{R})$.

Proof: We prove by induction over n. Consider the following cases:

1. $n = 0$. Then we have $D_0^{m+1}(\mathbb{R}) = D_0^n(\mathbb{R}) = \mathbb{R}$, and the statement holds;
2. $m = 0$. Then we have

$$D_n^1(\mathbb{R}) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^0(\mathbb{R})) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R})) = C_n(\mathbb{R}) = D_n^0(\mathbb{R}),$$

   and the inclusion is continuous by the definition of differential topology.
3. $n, m > 0$, with the statement proven for $(m, n-1)$ and $(m-1, n)$. Let $f \in D_n^{m+1}(\mathbb{R})$. Then we have

$$f \in \mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \subset \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})).$$

   Moreover, since $f' \in D_n^m(\mathbb{R}) \subset D_n^{m-1}(\mathbb{R})$, we see that $f \in D_n^m(\mathbb{R})$ by definition, so we have the inclusion $D_n^{m+1}(\mathbb{R}) \subset D_n^m(\mathbb{R})$. It only remains to show that the inclusion map $j : D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$ is continuous. This is equivalent to showing that the maps $i \circ j$ and $d \circ j$ are continuous, which is evident from the following commutative diagrams:

$$\begin{tikzcd}
D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, hook, "i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathcal{C}\big(\mathbb{R}, D_{n-1}^{m+1}(\mathbb{R})\big) \arrow[r, hook, "j"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))
\end{tikzcd}
\qquad
\begin{tikzcd}
D_n^{m+1}(\mathbb{R}) \arrow[r, hook, "j"] \arrow[d, "d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\
D_n^m(\mathbb{R}) \arrow[r, hook, "j"'] & D_n^{m-1}(\mathbb{R})
\end{tikzcd}$$

■

Corollary 3.1.1. *We have a chain of subspaces with each topology finer than the previous one:*

$$C_n(\mathbb{R}) = D_n^0(\mathbb{R}) \xleftarrow{j} D_n^1(\mathbb{R}) \xleftarrow{j} D_n^2(\mathbb{R}) \xleftarrow{j} D_n^3(\mathbb{R}) \xleftarrow{j} \dots$$

Lemma 3.1.2. *For all $n, m \in \mathbb{N}_0$, the topological space $D_n^m(\mathbb{R})$ is Hausdorff.*

Proof: Since $C_n(\mathbb{R})$ is Hausdorff and $i : D_n^m(\mathbb{R}) \to C_n(\mathbb{R})$ is continuous and injective, we see that $D_n(\mathbb{R})$ is also Hausdorff. ■

Lemma 3.1.3. *For all $m, n \in \mathbb{N}_0$, the pointwise linear structure is compatible with the topology on $D_n^m(\mathbb{R})$, making $D_n^m(\mathbb{R})$ a topological vector space.*

Proof: We prove by induction over n and m. If $n = 0$, we have $D_0(\mathbb{R}) = \mathbb{R}$ with the usual topology and linear structure. If $m = 0$, we have $D_n^0(\mathbb{R}) = C_n(\mathbb{R})$, which is a TVS. Suppose now that $m, n > 0$. We need to establish the continuity of the following two maps:

$$+ : D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}), \qquad \cdot : \mathbb{R} \times D_n^m(\mathbb{R}) \to D_n^m(\mathbb{R}).$$

In turn, by the definition of the differential topology, this task is equivalent to proving that the maps $i \circ (+), d \circ (+), i \circ (\cdot), d \circ (\cdot)$ are continuous, where the maps i and d are defined in (2).

To this end, note that we have commutative diagrams

$$\begin{tikzcd} D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, hook, "i \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\ \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(+)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \end{tikzcd}$$

$$\begin{tikzcd} D_n^m(\mathbb{R}) \times D_n^m(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, "d \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\ D_n^{m-1}(\mathbb{R}) \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(+)"'] & D_n^{m-1}(\mathbb{R}) \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, hook, "\mathrm{id} \times i"'] & D_n^m(\mathbb{R}) \arrow[d, hook, "i"] \\ \mathbb{R} \times \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \arrow[r, "(\cdot)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})) \end{tikzcd}$$

$$\begin{tikzcd} \mathbb{R} \times D_n^m(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, "\mathrm{id} \times d"'] & D_n^m(\mathbb{R}) \arrow[d, "d"] \\ \mathbb{R} \times D_n^{m-1}(\mathbb{R}) \arrow[r, "(\cdot)"'] & D_n^{m-1}(\mathbb{R}) \end{tikzcd}$$

Since the down-then-right path in each diagram is continuous, the right-then-down paths are continuous as well, and we are done. ∎

Theorem 3.1.1. *Let $m, n \in \mathbb{N}$. A curried function $f \in H_n(\mathbb{R})$ lies in the class $D_n^m(\mathbb{R})$ if and only if its counterpart $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives on $\mathbb{R}^n$, of all orders up to m. Moreover, for each $1 \le i_1 \le i_2 \le ... \le i_m \le n$, we have*

$$\begin{aligned} &\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}(x_1, x_2, ..., x_n) \\ &= f(x_1)...\left(x_{i_1-1}\right)'\left(x_{i_1}\right)...\left(x_{i_2-1}\right)'\left(x_{i_2}\right)...\left(x_{i_m-1}\right)'\left(x_{i_m}\right)...(x_n). \end{aligned} \tag{3}$$

Proof: If $n = 0$ or $m = 0$, the statement is trivial. Now let $n, m > 0$ and assume the statement proven for $(m, n-1)$ and $(m-1, n)$.

First, let $f \in D_n^m(\mathbb{R})$. We immediately see that $f \in D_n^{m-1}(\mathbb{R})$, and so, by the induction hypothesis, partial derivatives or order up to $m-1$ exist and are continuous on $\mathbb{R}^n$. Now, consider some $1 \le i_1 \le i_2 \le ..., \le i_m \le n$. It is not hard to see by definition that (3) holds. To show that the partial derivative with respect to $x_{i_1}, ..., x_{i_m}$ is continuous, consider $(h_1, h_2, ..., h_n) \to 0 \in \mathbb{R}^n$, and a point $(x_1, ..., x_n) \in \mathbb{R}^n$. By the continuity of f and the definition of the differential topology, we have

$$
\begin{aligned}
f(x_1 + h_1) &\xrightarrow[h_1 \to 0]{} f(x_1) \in D^m_{n-1}(\mathbb{R}),\\
&\vdots\\
f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)' &\xrightarrow[h_1,...,h_{i_1-1} \to 0]{} f(x_1)...\left(x_{i_1-1}\right)' \in D^{m-1}_{n-i_1+1}(\mathbb{R}),\\
&\vdots\\
&f(x_1 + h_1)...\left(x_{i_1-1} + h_{i_1-1}\right)'...\left(x_{i_m-1} + h_{i_m-1}\right)...(x_n + h_n)\\
&\xrightarrow[h_1,h_2,...,h_n \to 0]{} f(x_1)...\left(x_{i_1-1}\right)'...\left(x_{i_m-1}\right)'...(x_n) \in D^0_0(\mathbb{R}) = \mathbb{R},
\end{aligned}
$$

so the partial derivative

$$
\frac{\partial^m \tilde{f}}{\partial x_{i_1} \partial x_{i_2} ... \partial x_{i_m}}
$$

is continuous on $\mathbb{R}^n$.

Conversely, suppose $f \in H_n(\mathbb{R})$ is such that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ lies in the class $C^m(\mathbb{R}^n)$. To show that $f \in D^m_n(\mathbb{R})$, we follow three steps:

1. For all $x_1 \in \mathbb{R}$, we have $f(x_1) \in D^m_{n-1}(\mathbb{R})$. This is provided by the induction hypothesis, since the restriction of a C^m function on a hyperplane is also a C^m function.
2. The function $f : \mathbb{R} \to D^m_{n-1}(\mathbb{R})$ is continuous. To show this, consider a sequence $\left\{x_k^{(1)}\right\}_{k \in \mathbb{N}} \subset \mathbb{R}$ converging to $x_0^{(1)} \in \mathbb{R}$. We aim to prove that $f\left(x_k^{(1)}\right) \to f\left(x_0^{(1)}\right)$ in $D^m_{n-1}(\mathbb{R})$. This is equivalent to showing that

$$
f\left(x_k^{(1)}\right) \to f\left(x_k^{(1)}\right) \in \mathcal{C}(\mathbb{R}, D^m_{n-2}(\mathbb{R})) \quad \text{and} \quad f\left(x_k^{(1)}\right)' \to f\left(x_0^{(1)}\right)' \in D^{m-1}_{n-1}(\mathbb{R}). \tag{4}
$$

   The latter of these conditions holds by the induction hypothesis, since

$$
\frac{\partial \tilde{f}}{\partial x_2}(x_1, ..., x_n) = \frac{\partial \widetilde{f(x_1)}}{\partial x_2}(x_2, ..., x_n) \underset{\text{by (3)}}{=} f(x_1)'(x_2)...(x_n), \tag{5}
$$

   and $\partial \tilde{f} / \partial x_2$ is continuous, which means that the map $x \mapsto f(x)'$ is continuous. To prove the former condition in (4), we again consider a sequence $x_k^{(2)} \to x_0^{(2)} \in \mathbb{R}$ and aim to prove

$$
f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right) \in D_{n-2}(\mathbb{R}),
$$

   which (unless $n = 2$) is again equivalent to

$$
f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \quad \text{and} \quad f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)' \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)'.
$$

   The latter follows from the induction hypothesis and the continuity of $\partial \tilde{f} / \partial x_3$, similarly to (5), and the former can again be expanded further. This chain leads us finally to the convergence

$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)...\left(x_k^{(n)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right) \in \mathbb{R},$$

which is evident since $f \in C_n(\mathbb{R})$ by Proposition 3.1. Hence we see that f lies in $\mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))$. We have also established (3) for f and $i \geq 2$, via (5).

3. Finally, we need to find $f' \in D_n^{m-1}(\mathbb{R})$ such that

$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h\to 0]{} f'(x) \in H_{n-1}(\mathbb{R}) \tag{6}$$

for all $x \in \mathbb{R}$. Indeed, following (3), let us define $f' := \gamma\left(\partial\tilde{f}/\partial x_1\right)$. We immediately see by the induction hypothesis that f' so defined lies in $D_n^{m-1}(\mathbb{R})$. To see that (6) holds, consider $x_1, ..., x_n \in \mathbb{R}$ and write

$$\begin{aligned} f'(x_1)(x_2)...(x_n) &= \frac{\partial \tilde{f}}{\partial x_1}(x_1, ..., x_n) \\ &= \lim_{h\to 0} \frac{\tilde{f}(x_1 + h, x_2, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} \\ &= \left(\lim_{h\to 0} \frac{f(x_1 + h) - f(x_1)}{h}\right)(x_2)...(x_n), \end{aligned}$$

and we are done.

■

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

[2] N. Bourbaki, *General Topology, Part II*. 1966.

[3] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.

4. Appendix

4.1. The language of filters

Definition 4.1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 4.1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 4.1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 4.1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Lemma 4.1.1. (see [1], page 74): *Let X be a set and $f_i : X \to Y_i$ be a family of maps, where Y_i are topological spaces. Let X be endowed with the coarsest topology such that f_i is continuous for each i. Then a filter $F \in \mathcal{F}(X)$ converges to $x \in X$ if and only if $f_i(F)$ converges to $f_i(x)$ for all i.*

4.2. Topological vector spaces

We assume that the reader is familiar with the definition and basic properties of topological vector spaces. Here we list some results that will be

4.3. The topology of pointwise convergence

Definition 4.3.1. (pointwise convergence topology, see [2]): Let X be a set, Y a topological space, and denote by $\mathcal{S}(X, Y)$ the set of functions from X to Y. For each $x \in X$ and each open set $U \subset Y$, let $V(x, U)$ be the set of all functions $f \in \mathcal{S}(X, Y)$ such that $f(x) \in U$. The collection
$$\{V(x, U) \mid x \in X, U \subset Y\}$$
forms a subbase of a topology on $\mathcal{S}(X, Y)$, called the *topology of pointwise convergence.*

Remark 4.3.1. Let X be a set and Y a topological space. It is not hard to see that a filter $F \in \mathcal{F}(\mathcal{S}(X, Y))$ converges to a function $f \in \mathcal{S}(X, Y)$ if and only if, for any $x \in X$, the filter $F(x) = [\{A(x) \mid A \in F\}]$ converges to $f(x) \in Y$.

Lemma 4.3.1. *Let X be a set and Y a Hausdorff topological space. Then $\mathcal{S}(X, Y)$ is also Hausdorff.*

> *Proof:* Consider two functions $f, g \in \mathcal{S}(X, Y)$ such that $f \neq g$. This implies that there is $x_0 \in X$ such that $f(x_0) \neq g(x_0)$. Since Y is Hausdorff, there are neighborhoods $U_f \ni f(x_0)$ and $U_g \ni g(x_0)$ such that $U_f \cap U_g = \varnothing$. It then follows that $f \in V(x_0, U_f)$, $g \in V(x_0, U_g)$, and $V(x_0, U_f) \cap V(x_0, U_g) = \varnothing$. ■

Lemma 4.3.2. *Let X be a set and let Y be a real topological vector space. Then $\mathcal{S}(X, Y)$ is also a real topological vector space with respect to the topology of pointwise convergence and the pointwise linear structure.*

> *Proof:* We need to show that the maps $+ : \mathcal{S}(X, Y) \times \mathcal{S}(X, Y) \to \mathcal{S}(X, Y)$ and $\cdot : \mathbb{R} \times \mathcal{S}(X, Y) \to \mathcal{S}(X, Y)$ are continuous.

To begin, consider two filters $F_1, F_2 \in \mathcal{F}(\mathcal{S}(X,Y))$, converging to f_1 and f_2 respectively. To show that $F_1 + F_2$ converges to $f_1 + f_2$, consider $x \in X$. We have

$$\begin{aligned}(F_1 + F_2)(x) &= [\{(A+B)(x) \mid A \in F_1, B \in F_2\}] \\ &= [\{A(x) + B(x) \mid A \in F_1, B \in F_2\}] \\ &= F_1(x) + F_2(x) \to f_1(x) + f_2(x) = (f_1 + f_2)(x).\end{aligned}$$

This shows that the addition operation is continuous. Scalar multiplication is continuous by a similar argument. ■

4.4. The compact-open topology

Definition 4.4.1. (compact-open topology, see [2]): Let X and Y be two topological spaces, and by $\mathcal{C}(X,Y)$ denote the set of all continuous functions from X to Y. For each compact set $K \subset X$ and each open set $U \subset Y$, let $V(K,U)$ be the set of all functions $f \in \mathcal{C}(X,Y)$ such that $f(K) \subset U$. The collection

$$\{V(K,U) \,|\, K \subset X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{C}(X,Y)$, called the *compact-open* topology.

Remark 4.4.1. Clearly, the topology of pointwise convergence on $\mathcal{C}(X,Y)$ is coarser than the compact-open topology.

Proposition 4.4.1. (see [3]): *Let X and Y be topological spaces such that X is locally compact and Y is regular. Then a filter $F \in \mathcal{F}(\mathcal{C}(X,Y))$ converges to a function $f \in \mathcal{C}(X,Y)$ if and only if, for any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have $F(P) \to f(p)$ in Y, where the filter $F(P)$ is based on*

$$\{A(B) \,|\, A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \,|\, A \in F, B \in P\}.$$

Proposition 4.4.2. (see [2], page 302): *Let X, Y, Z be topological spaces such that X is Hausdorff and Y is locally compact. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y,Z))$$

defined as $\gamma(f)(x)(y) = f(x,y)$, is well-defined and is a homeomorphism.

Lemma 4.4.1. *Let X, Y be topological spaces such that Y is Hausdorff. Then the space $\mathcal{C}(X,Y)$ is also Hausdorff.*

Proof: The compact-open topology on $\mathcal{C}(X,Y)$ is finer than the topology of pointwise convergence, and the latter is Hausdorff, which implies that the former is Hausdorff. ■

Lemma 4.4.2. *Let X be a topological space and Y a real topological vector space. Then $\mathcal{C}(X,Y)$ is also a real topological vector space with respect to the pointwise linear structure and the compact-open topology.*

Proof: We will show that the map $+ : \mathcal{C}(X,Y) \times \mathcal{C}(X,Y) \to \mathcal{C}(X,Y)$ is continuous. Let $F_1, F_2 \in \mathcal{F}(\mathcal{C}(X,Y))$ converge to $f_1 \in \mathcal{C}(X,Y)$ and $f_2 \in \mathcal{C}(X,Y)$ respectively. For any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have

$$
\begin{aligned}
(F_1 + F_2)(P) &= [\{(A_1 + A_2)(B) \mid A_i \in F_i, B \in P\}] \\
&\supset [\{A_1(B_1) + A_2(B_2) \mid A_i \in F_i, B_i \in P\}] \\
&= F_1(P) + F_2(P) \to f_1(p) + f_2(p) = (f_1 + f_2)(p).
\end{aligned}
$$

The continuity of scalar multiplication is proven similarly. ■